PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—June PAPER 1—

2 June 2025

Examiner: Mr. A Meintjes
Internal Moderator: Mr. D. Grace

TIME: 3 hours **MARKS: 150 Marks**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. Write your name and class in the space provided on the answer book.
2. Please check that your question paper is complete.
3. Read the questions carefully and answer ALL the questions in your answer booklet.
4. It is in your own interest to write legibly and to present your work neatly.
5. You are provided with a 1 : 50 000 topographic map 2926AA Bloemfontein (NORTH) topographical map and the 2926 10 Bloemfontein orthophoto.
6. You must hand the topographic map and orthophoto map to the invigilator at the end of this examination session. You may only use pencil on these maps. Please erase any pencil markings on the map before handing it back in.
7. Show ALL calculations and formulae, where applicable. Marks will be allocated for these.
8. Indicate the unit of measurement or compass direction in the final answer of calculations, e.g. 10 km; 2.1cm; west of true north.
9. You may use a non-programmable calculator.
10. You may make use of a magnifying glass.
11. The area demarcated in RED/BLACK on the topographic map represents the area covered by the orthophoto map.

SECTION A – Climate and Weather

QUESTION 1: Short Questions

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Cells of low pressure can be defined as …

A. A localised elongated line of low pressure such as a cold front.
B. Areas of differing pressure that extend around the planet going from East to West.
C. Concentrated areas of low pressure often resulting in thunderstorms.
D. Areas of differing pressure that extend around the planet going from North to South. (1)

1.1.2. A South-facing slope in South Africa would experience ...

A. Warmer conditions between December and February.
B. Colder conditions between December and February.
C. Consistently warmer conditions throughout the year.
D. Consistently cooler conditions throughout the year. (1)

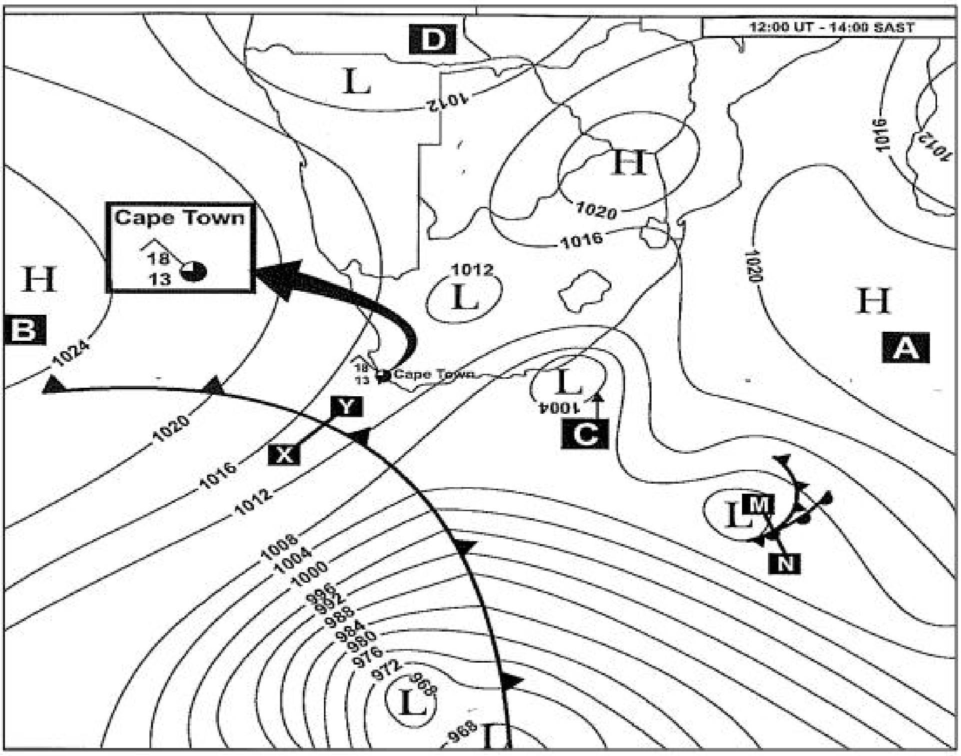
1.1.3. Conditions necessary for frost pockets to form include …

A. Clear skies and cold temperatures.
B. No strong winds at all.
C. Light rain and no strong winds over the valley.
D. Only A and B. (1)

1.1.4. Anabatic winds refer to …

A. Winds that blow up valley sides and along the valley floor during the day.
B. Winds that travel vertically up in cities due to excess heat sources.
C. Sinking winds resulting from temperature inversions.
D. Winds that blow down valley sides and along the valley floor during the night. (1)

Refer to the diagram below showing a typical synoptic weather map and answer the questions that follow.



1.1.5. The season shown above is most likely …

A. Summer.

B. Autumn.

C. Winter.

D. Spring. (1)

1.1.6. The weather phenomenon labelled C is a …

A. Mid-latitude cyclone.

B. Coastal low pressure.

C. Tropical cyclone.

D. Cut-off low. (1)

1.1.7. The high pressure cell dominating the interior of South Africa has developed because …

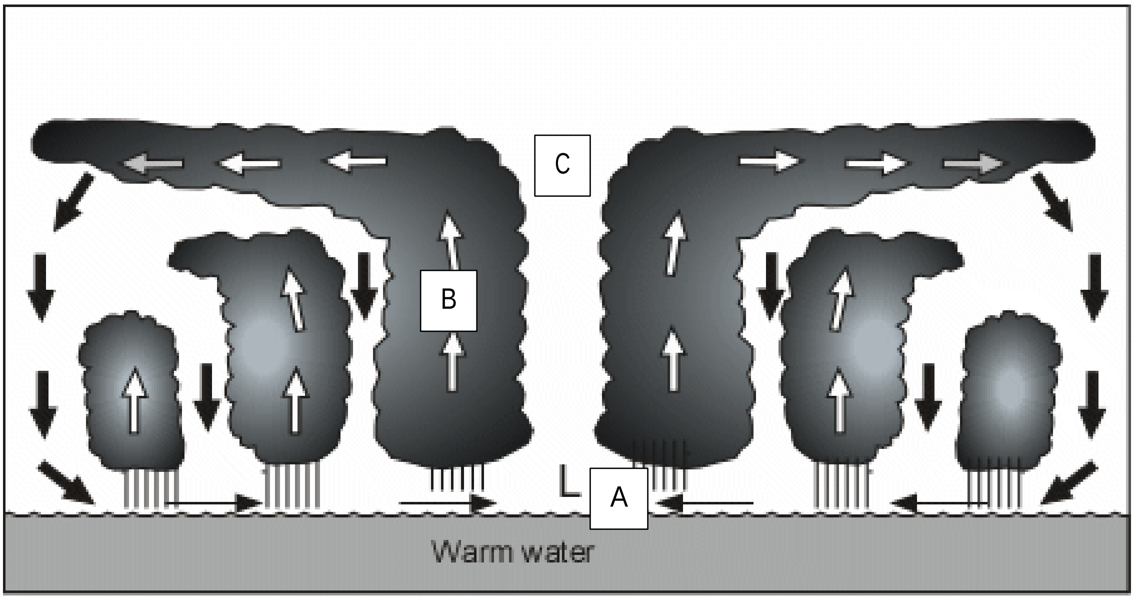
i. There is a decrease in heat this time of the year.
ii. The Kalahari High Pressure has increased.
iii. The interior is cooler than the coastline.
iv. The cold Benguela current has strengthened.

A. i & iii ONLY.
B. None of the above.
C. All of the above.
D. i, ii & iii ONLY. (1)

1.1.8. Low pressure systems in **Summer** can lead to …
A. Colder conditions over a few weeks.
B. Clearer skies.
C. Increased flooding risks.
D. Increased temperature over a few days. (1)

(8 x 1) (8)

1.2. Refer to the diagram below and answer the questions that follow. For each answer select the correct word in brackets. Write only the question and the correct word. E.g. 1.2.8 Jam.



1.2.1. As the cyclone approaches we would experience (warm temperatures and high humidity/low temperatures and low humidity).
1.2.2. The eye of the cyclone has the (most powerful winds/calmest winds).
1.2.3. The clouds found at B are (alto-nimbus/cumulonimbus) clouds.
1.2.4. The air movement at C is (descending/rising).
1.2.5. The ocean surface at A will experience a (stormsurge/tidal wave).
1.2.6. In order for tropical cyclones to form, they need Coriolis force, hot moist air and intense (high/low) pressure cells forming above.
1.2.7. These massive storms typically form towards the (beginning/end) of summer when the ocean is at its warmest.

(7 x 1) (7)

[15]

QUESTION 2: Mid-latitude Cyclones

2. Refer to the news article below and answer the questions that follow.

Bomb cyclone? Land hurricane? Powerful storm heading our way evokes interesting names

by FOX 11 Meteorologist Phil DeCastro
Wed, March 12th 2025 at 4:48 PM

Updated Thu, March 13th 2025 at 7:27 AM

At every transitional season, the spring and the fall, powerful mid-latitude cyclones tend to sweep across the United States.

And with the most powerful ones, you start hearing terms like "bomb cyclone" -- or people even likening the storms to hurricanes.

Some of these terms are legit -- some aren't. Let's start with my favorite: the "land hurricane".

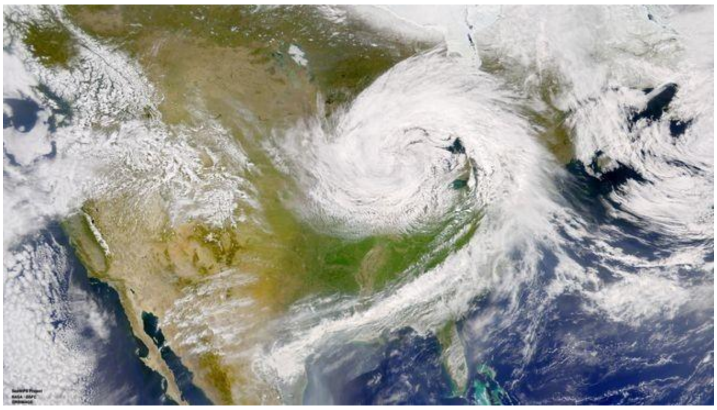
First of all, hurricanes form due to a completely different process than storms like the one tracking across the U.S. late this week. Hurricanes are a warm-core cyclone, driven by a warm ocean and the hot, humid air atop it rising and forming thunderstorms, and the air at their center tends to be warmer than the environment around it.

Storms like the one trekking over land across the U.S. later this week are driven by big horizontal differences in air temperature, and as they strengthen, the air at the center ends up being colder than the environment around it.

They're called cold-core cyclones. So in that regard, this is not going to be a "land hurricane." But I will grant two things.

One, the severe thunderstorms triggered by these mid-latitude cyclones can absolutely produce straight-line winds of hurricane strength, which is anything above 74 miles per hour.

Two, they can sometimes kinda-sorta look alike. It's not perfect, but they both can take on swirling, rotating shapes when viewed by satellite.



Cold-core cyclones, though, do not develop eyes at their center like hurricanes do.

The other term you might hear is "bomb cyclone" and this one is easy enough to tackle. When the central pressure of a cyclone decreases by 24 millibars in 24 hours, it's considered to be "bombing out" or undergoing "bombogenesis".That is, the pressure at the center of the storm is dropping like a bomb. Or, you could say its strength is exploding, like a bomb.

Either way, when you hear a system referred to as a "bomb cyclone" you know it's going to be a powerful one.

2.1. Which direction do mid-latitude cyclones travel in the northern hemisphere, west to east or east to west? (1 x 1) (1)

2.2. According to the article, what are the TWO key differences between mid-latitude cyclones and hurricanes? (2 x 2) (4)

2.3. What is a 'bomb cyclone' according to the article? (1 x 2) (2)

2.4. Differentiate between the Mature Stage and Occulsion Stage of mid-latitude cyclone development. (2 x 2) (4)

2.5. Draw a simple cross section of a cold front occlusion. Include the following:

A heading for your diagram

Correct front shapes and symbols

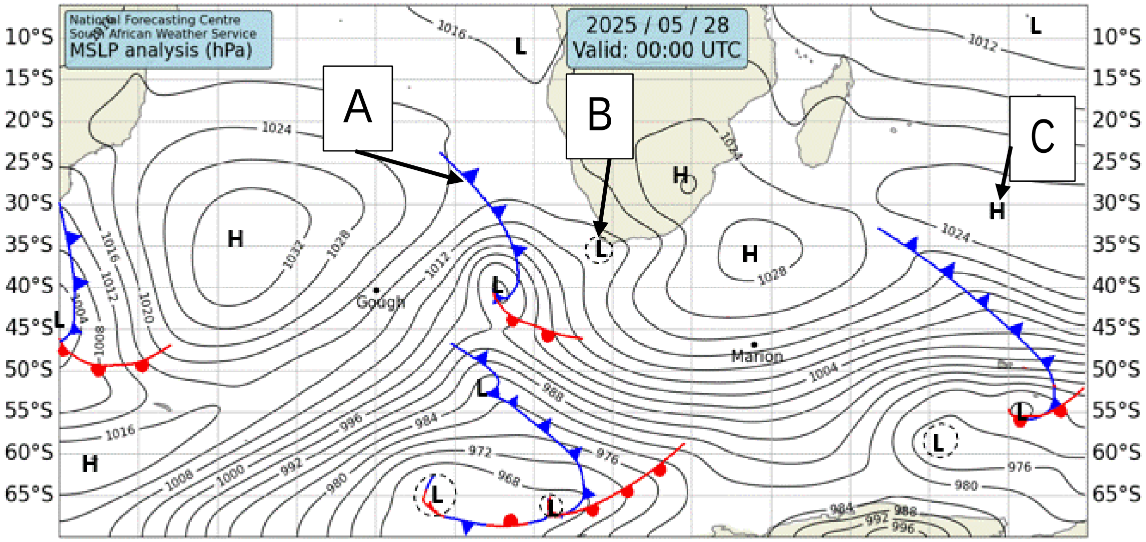
Correct air pocket temperatures

Correct cloud types (4 x 1) (4)

[15]

QUESTION 3: Anti-cyclones

3. Study the synoptic weather map below and answer the questions that follow.



3.1. What season is this synoptic weather map depicting? (1 x 1) (1)

3.2. Provide ONE piece of evidence from the chart to justify your answer to QUESTION 3.1. (1 x 1) (1)

3.3. Provide the names for the features labelled A and C on the map respectively. (2 x 1) (2)

3.4. The system labelled B is a coastal low pressure and therefore a travelling disturbance. Which way is it likely to travel, east or west? (1 x 1) (1)

3.5. Draw a simple cross section showing how the low pressure named at B may lead to the formation of Berg Winds. (4 x 1) (4)

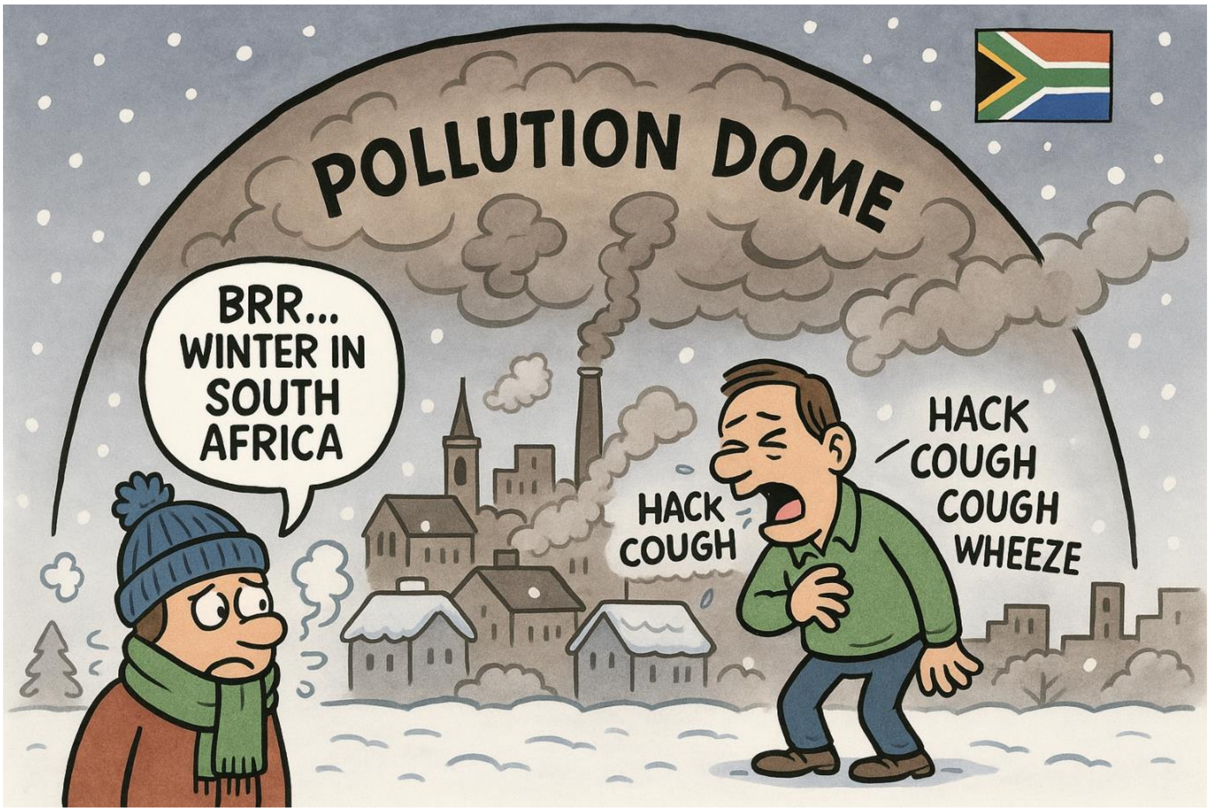
3.6. Discuss why Berg Winds are a danger to South Africa. Provide TWO reasons in your answer (2 x 2) (4)

3.7. Describe the role that the Kalahari High Pressure cell plays in the reduction of rain over the interior during this season. (1 x 2) (2)

[15]

QUESTION 4: Urban Climates

4. Study the ChatGPT image and answer the questions that follow.



4.1. Did ChatGPT get the season that pollution domes form in right? (1 x 1) (1)

4.2. What is the season depicted in this image? (1 x 1) (1)

4.3. What characteristic of South African cities did ChatGPT get wrong? (1 x 1) (1)

4.4. Describe TWO factors that lead to the formation of pollution domes. (2 x 2) (4)

4.5. In a short paragraph, briefly discuss TWO negative economic impacts of pollution domes and TWO solutions that local governments could implement to reduce these impacts. (2 x 4) (8)

[15]

TOTAL SECTION A: [60]

SECTION B - Geomorphology

QUESTION 5: Short Questions

5.1. The following descriptions are all features of a river system. State whether each of these statements is TRUE or FALSE. Write only the question and the answer. E.g. 5.1.8. True

5.1.1. Deposition is dominant at in the lower course of a river.

5.1.2. A transverse river profile shows a general path of a river from the source to the mouth.

5.1.3. Meandering rivers and ox-bow lakes are a features of the upper river course.

5.1.4. Most headward erosion takes place in the middle course of a river.

5.1.5. Oxbow lakes typically form within deltas.

5.1.6. The Nile travelling through the desert would be an example of an episodic river.

5.1.7. Debris in a river moving through traction refers to the process whereby large boulders are removed with machines.

(7 x 1) (7)

5.2. Match the terms in the table with one of the descriptions. Write only the number of the question and the correct term (5.2.1–5.2.8) in the ANSWER BOOK, for example 5.2.9 Sunshine.

Catchment area	**Knickpoint**	**Sinuosity**
Valley	**Meander**	**Abrasion**
Antecedent drainage	**Hydraulic action**	**Source**
Watershed	**Confluence**	**Interfluves**
Mouth	**Laminar flow**	**Turbulent flow**

5.2.1. This is the point where one river joins another river.

5.2.2. High ground separating one drainage basin from another adjacent drainage basin.

5.2.3. A ridge or area of land dividing two river valleys within a drainage basin.

5.2.4. A term to describe a particular point in a river or channel where there is a sharp drop in altitude, usually resulting in the formation of waterfalls.

5.2.5. A process whereby the continual colliding of water and other materials on the base of the plunge pool gradually deepens the pool.

5.2.6. The term that refers to the end of a river.

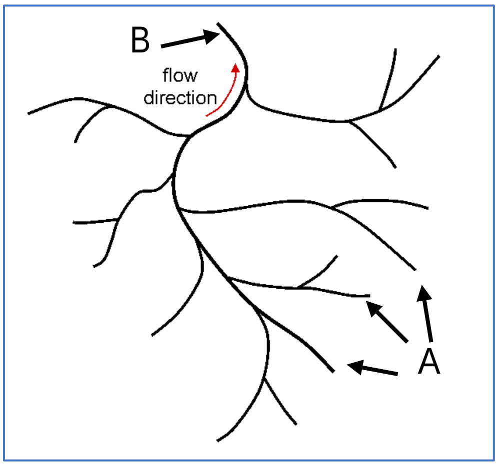
5.2.7. The area from which rainfall flows into a river, lake, or reservoir.

5.2.8. A term used to describe calm water with little turbulence.

(8 x 1) (8)
[15]

QUESTION 6: Drainage basins

6. Refer to the figure below showing a drainage basin and answer the questions that follow.



6.1. Provide labels for river features at A and B respectively? (2 x 1) (2)

6.2. What is the shape of this drainage basin? (1 x 1) (1)

6.3. Determine the stream order of the river exiting at B. (1 x 2) (2)

6.4. Does this shape indicate that this basin *depends* on the geology and topography of the area, or is it *independent* of the geology and topography of the area? (1 x 1) (1)

6.5. Using the below information, calculate the drainage density of this basin.

Length of streams = 44.5km

Area of basin = 8.23km²

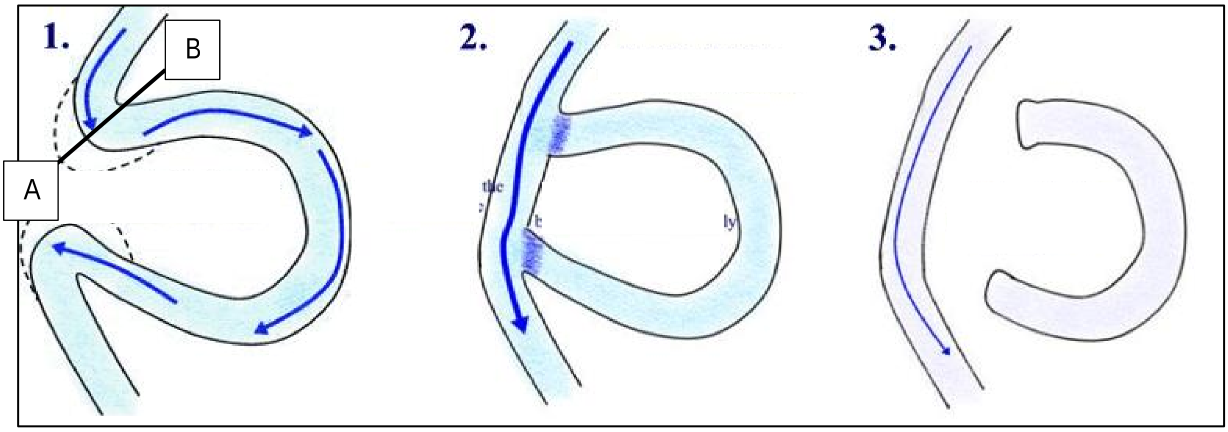
Formula: $\frac{Total\ length\ of\ streams\ in\ basin\ (km)}{Total\ area\ of\ the\ drainage\ basin\ (km^2)}$ (3 x 1) (3)

6.6. There are many factors that can influence drainage density. Briefly discuss any THREE of these factors. (3 x 2) (6)

[15]

QUESTION 7: Fluvial Processes

7. Study the figure below and answer the questions that follow.



7.1. What feature is being formed in the diagram above? (1 x 1) (1)

7.2. In which course of the river are these features most likely to form? (1 x 1) (1)

7.3. Would the likely gradient for the above diagram be steep or gentle? (1 x 1) (1)

7.4. In diagram 1 the river is meandering. Draw a simple cross section between points A and B showing how the water is flowing in this part of the river. Include the following:

Flow speeds (indicate the strongest flow and slowest flow)

Deposition (point bars)

Erosion (undercutting) (4 x 1) (4)

7.5. In diagram 1, explain the process occurring at the "narrow neck" of the meander. (1 x 2) (2)

7.6. Features like the above often develop as a result of river rejuvenation. Define the term *River Rejuvenation.* (1 x 2) (2)

7.7. Describe TWO negative consequences for people and/or animals that live downstream of a river that has been rejuvenated. (2 x 2) (4)

[15]

QUESTION 8: River Management

Study the article below and answer the questions that follow.

One river running through South Africa's richest province is so polluted it is killing people

Once a shimmering centerpiece of the area, Centurion Lake in Gauteng has become, in the words of some local residents, "a pungent ghost town that nobody wants to go near."

Standing on the banks of what was once a bustling tourist attraction, with vibrant restaurants and shops, the overwhelming smell of raw sewage now dominates the scene.

Lake Centurion is now so filled with sediment and pollution that it no longer functions as a proper lake. Incoming water from the Hennops River is heavily contaminated with sewage, plastic, and toxins.

Water expert Professor Anthony Turton told BusinessTech that Hennops pollution has been a persistent problem, driven by industrial runoff, long-standing sewage discharge, and poor wastewater management.

As a result, adjacent properties like Centurion Mall have suffered falling real estate values and tenant losses.

To date, the river continues to carry its pollution toward Hartbeespoort Dam, amid controversial claims of radionuclide contamination near Pelindaba, which authorities have denied.

Illegal sand mining, especially upstream, has exacerbated sediment buildup, contributing to the lake's degradation and reflecting a broader national problem.

On top of this, densely populated surrounding areas like Tembisa, Ivory Park, and Olifantsfontein lack essential services such as sanitation, waste removal, and stormwater control.

As a result, the Hennops is continuously burdened with raw sewage, industrial effluent, and litter.

State of the river

A major pressure point is the Olifantsfontein Wastewater Treatment Works (WWTW). Designed to handle 110 million litres per day, it is routinely overwhelmed, sometimes processing nearly double its capacity.

This results in tens of millions of litres of untreated or poorly treated effluent flowing into the river daily.

"The simple truth is that WWTWs are slowly rendering the drinking water of the country unusable," said Turton.

"This is now a material factor in the dysfunction of the Ekurhuleni Water Care Company plant, along with fats, oil and grease, and of all things, clogging of the machinery," wrote Turton for the Daily Maverick.

Very broadly, these issues have made the Hennops River now one of the most polluted waterways in Gauteng.

A recent study by Thabiso Letseka from the University of the Witwatersrand's Faculty of Science confirmed many fears about the dangers of the river, finding it critically polluted and unfit for drinking, irrigation, or recreation.

The South African Human Rights Commission (SAHRC) found that local mismanagement of wastewater treatment plants has been key in the severe pollution of rivers like the Hennops, threatening ecosystems, health, and livelihoods.

Despite decades of complaints and directives, the SAHRC said that the government has failed to act, violating constitutional rights to a safe environment and proper water services.

"Without sounding dramatic, South Africa is slowly committing ecocide – national suicide by poisoning its own drinking and crop production water," Turton previously told BusinessTech.

8.1. In which province is Centurion Lake located? (1 x 1) (1)

8.2. What is the main source of pollution in the Hennops River? (1 x 1) (1)

8.3. List TWO economic impacts the pollution of Centurion Lake has had on the local economy. (2 x 1) (2)

8.4. How does poor service delivery in areas like Tembisa and Ivory

Park affect the health of the Hennops River? (1 x 2) (2)

8.5. Which constitutional right is being violated according to the SAHRC? (1 x 1) (1)

8.6. Considering the following quote from the article: "Without sounding dramatic, South Africa is slowly committing ecocide – national suicide by poisoning its own drinking and crop production water." Write a short paragraph in which you explore what the future may hold for South Africans if we do not improve our current situation, and provide solutions that leadership could consider to improve the situation. (4 x 2) (8)

[15]

TOTAL SECTION B: [60]

SECTION C - Mapwork

RESOURCE MATERIAL

1. 2926AA Bloemfontein (NORTH) topographical map and the 2926 10 Bloemfontein orthophoto

GENERAL INFORMATION ON Bloemfontein

Bloemfontein (/ˈbluːmfɒnteɪn/ *BLOOM-fon-tayn*;[4][5] Afrikaans: [ˈblumfɔntəin]), also known as **Bloem**, is the capital and the largest city of the Free State province in South Africa. It is often, and has been traditionally, referred to as the country's "judicial capital", alongside the legislative capital Cape Town and administrative capital Pretoria, although the highest court in South Africa, the Constitutional Court, has been in Johannesburg[6][7][8] since 1994.

Situated at an elevation of 1,395 m (4,577 ft) above sea level, the city is home to 256,185 (as of 2011)[9] residents and forms part of the Mangaung Metropolitan Municipality which has a population of 747,431.[10] It was one of the host cities for the 2010 FIFA World Cup.

QUESTION 9: Calculations

9.1. What is located at F on the topographical map? (1 x 1) (1)

9.2. Provide the coordinates for communication tower in E1. (2 x 1) (2)

9.3. Provide the map code or map reference for the map North of 2926AA. (1 x 1) (1)

9.4. Choose the correct option below.

The area covered by the orthophoto map as represented on the topographic map is approximately:

A) 3145km².
B) 3.145km².
C) 3.145m².
D) 3145m².

(1 x 2) (2)

9.5. Calculate the current magnetic declination for 2025. Show ALL calculations. Use the following headings as a guide:

The difference in years:
Mean annual change:
Total change:
Magnetic declination for 2025:

(4 x 1) (4)

[10]

QUESTION 10: Interpretation

10.1. Refer to the river network in B1.

10.1.1. What type of rivers dominate this area, permanent or exotic? (1 x 1) (1)

10.1.2. What are the primary uses of these rivers? (1 x 1) (1)

10.1.3. This river system has a low drainage density. Discuss TWO factors that would influence this system's drainage density. (2 x 2) (4)

10.2. Refer to the vast amount of industrialisation in Tempe (block E2).

10.2.1. How may this industrialisation lead to acid rain in the surrounding areas in Winter? (1 x 2) (2)

10.2.2. During Summer the area of Tempe experiences a strong urban heat island effect. Draw a simple diagram showing the formation of an urban heat island. (2 x 1) (2)

10.2.3. Suggest ONE way that the people of Tempe can combat the effects of urban heat islands. (1 x 2) (2)

[12]

QUESTION 11: GIS

Refer to the Freestate National Botanical Gardens (B3 & C3) and the extract of information below from website and answer the questions that follow.

The status of biological invasions and their management in South Africa in 2022

11.1. Provide ONE example of line feature found within the Freestate National Botanical Gardens (Block B3 & C3). (1 x 1) (1)

11.2. Is the topographical map an example of raster or vector data? (1 x 1) (1)

11.3. Buffering was clearly used before building this park. Provide evidence from the map to justify this statement. (1 x 1) (1)

11.4. Consider the information provided above. How would GIS be used to monitor the pathways that invasive species move through? (1 x 2) (2)

11.5. When generating reports on these invasive species, would you use data manipulation or buffering when creating these charts? (1 x 1) (1)

11.6. Discuss how you would use GIS in the safeguarding of the water supply that runs through the gardens. (1 x 2) (2)

[8]

TOTAL SECTION C: [30]

<u>GRAND TOTAL: 150 Marks</u>